

D159

Evidence-Based Measures for Evaluating Healthcare Improvements

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A falls-prevention project on a 28-bed unit that recently switched EHR vendors tracks the percentage of patients who receive a fall-risk assessment within 4 hours of admission. The unit's manager has a doctorate in nursing practice. What type of measure is this?

- A. Balancing measure.
- B. Process measure.
- C. Outcome measure.
- D. Structure measure.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which is an outcome measure for a catheter-associated urinary tract infection (CAUTI) reduction project?

- A. Any measure reported to leadership.
- B. Percentage of catheters with a documented daily need review.
- C. Number of bladder scanners on the unit.
- D. CAUTIs per 1,000 catheter days.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A hospital with a new patient portal launches a project to cut average length of stay on its medical units from 5.2 to 4.5 days. The chief medical officer championed the idea at a board retreat. Which is the best balancing measure?

- A. 30-day readmission rate.
- B. Percentage of patients with a discharge plan by day 2.
- C. Number of patient portal sign-ups.
- D. Average length of stay.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

In the third quarter, a medical-surgical unit with 32 beds and a new charge-nurse model recorded 9 patient falls over 4,500 patient days. The previous quarter had 7 falls. What was the fall rate per 1,000 patient days?

- A. About 1.6
- B. 500
- C. 0.2
- D. 2.0

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

5. [TRANSFER — reversed stem / different variable]

A unit's goal is a fall rate no higher than 1.5 per 1,000 patient days. Next quarter it expects 6,000 patient days. What is the most falls it can have and still meet the goal?

- A. 4
- B. 1.5
- C. 9
- D. 90

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

6.

After a new hand-off tool was introduced by a popular clinical educator, a run chart of weekly medication-reconciliation errors shows the next 7 weekly points all below the baseline median. Two of the weeks had lower census because of a holiday. What does this pattern indicate?

- A. Random variation, since two weeks had low census.
- B. A shift, a non-random signal of change.
- C. A trend.
- D. Proof that the tool caused the change.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

7. [TRANSFER — reversed stem / different variable]

Which run chart pattern meets the rule for a trend?

- A.** Five consecutive points, each lower than the one before.
- B.** A single point far above all others.
- C.** Three consecutive points each higher than the last.
- D.** Six consecutive points above the median, going up and down.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A project team at a teaching hospital compares average ED door-to-provider time in minutes for 200 patients before a triage change with a different 200 patients after it. The team's statistician is a former high school math teacher. Which analysis fits best?

- A.** A chi-square test.
- B.** An independent-samples t-test comparing the two means.
- C.** No test; compare the two averages by eye.
- D.** A paired t-test.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A team wants to know whether the proportion of patients readmitted within 30 days (yes or no) differs between two discharge programs. Which analysis fits?

- A.** A chi-square test.
- B.** A correlation coefficient.
- C.** An independent-samples t-test.
- D.** A run chart only.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

To measure hand hygiene, a unit counts how many sanitizer pumps are used each shift. Two auditors who both trained at the same university agree on the counts 98% of the time. Staff admit some pumps are used to clean equipment. What is the best description of this measure?

- A. Valid because it is reliable.
- B. Valid but not reliable.
- C. Reliable but not valid.
- D. Both reliable and valid, since the auditors were trained together.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A project team measuring time to first antibiotic for sepsis asks IT for a data extract. The team lead, who has a new office near the cafeteria, requests every patient's full chart, including Social Security numbers and home addresses, "in case we need it later." What is the best response?

- A. Approve it if the extract is stored in the team lead's new office.
- B. Approve it, since quality improvement is a permitted use of health data.
- C. Limit the extract to the minimum necessary fields, such as timestamps and a study ID, and de-identify where possible.
- D. Refuse all data access for quality projects.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A central-line infection project on an ICU with a new family lounge will report its CLABSI rate quarterly. Because infections are rare, the nurse leader wants a weekly signal that the project is on track. The ICU has 20 beds. Which indicator best serves that purpose?

- A. Quarterly CLABSI rate.
- B. Number of beds in the ICU.
- C. Annual mortality rate for ICU patients.
- D. Weekly percentage of central lines with full maintenance-bundle compliance.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A falls-prevention project on a 28-bed unit ~~that recently switched EHR vendors~~ tracks the percentage of patients who receive a fall-risk assessment within 4 hours of admission. ~~The unit's manager has a doctorate in nursing practice.~~ What type of measure is this?

Target word: measure type **Decides:** It counts whether a care step was done.

A	Dropped element	Balancing measures track unintended effects elsewhere, not the main process.
B	Correct	It measures how reliably a care action is performed.
C	Confusion set	Outcomes are patient results such as the fall rate.
D	Adjacent category	Structure measures resources such as staffing or bed alarms.

2. Answer D Concept 1 • transfer

Which is an outcome measure for a catheter-associated urinary tract infection (CAUTI) reduction project?

Target word: reverse: pick the outcome **Decides:** Outcome = patient result.

A	Absolute language	Reporting audience does not decide the measure type.
B	Inverted operation	That is a process measure.
C	Adjacent category	Equipment is a structure measure.
D	Correct	Infection rate is a patient outcome.

3. Answer A Concept 2

A hospital ~~with a new patient portal~~ launches a project to cut average length of stay on its medical units from 5.2 to 4.5 days. ~~The chief medical officer championed the idea at a board retreat.~~ Which is the best balancing measure?

Target word: balancing measure **Decides:** Shorter stays risk earlier return to the hospital.

A	Correct	Readmissions could rise if patients leave too early.
B	Adjacent category	That is a process measure driving the change.
C	Camouflage	Unrelated to the risk created by shorter stays.
D	Dropped element	That is the outcome being improved, not a check on side effects.

4. Answer D Concept 3

In the third quarter, a medical-surgical unit ~~with 32 beds and a new charge nurse model~~ recorded 9 patient falls over 4,500 patient days. ~~The previous quarter had 7 falls.~~ What was the fall rate per 1,000 patient days?

Target word: fall rate **Decides:** $9 / 4,500 \times 1,000$.

A	Camouflage	Uses the previous quarter's 7 falls.
B	Inverted operation	Divides patient days by falls.
C	Wrong stem number	Multiplies by 100 instead of 1,000.
D	Correct	$9 / 4,500 = 0.002$; $\times 1,000 = 2.0$.

5. Answer C Concept 3 • transfer

A unit's goal is a fall rate no higher than 1.5 per 1,000 patient days. Next quarter it expects 6,000 patient days. What is the most falls it can have and still meet the goal?

Target word: solve for numerator **Decides:** Falls = $1.5 \times 6,000 / 1,000$.

A	Inverted operation	Divides 6,000 by 1,500 instead of multiplying the rate.
B	Dropped element	Reports the rate instead of the count.
C	Correct	$1.5 \times 6,000 / 1,000 = 9$.
D	Wrong stem number	Divides by 100 instead of 1,000.

6. Answer B Concept 4

After a new hand-off tool was ~~introduced by a popular clinical educator~~, a run chart of weekly medication-reconciliation errors shows the next 7 weekly points all below the baseline median. ~~Two of the weeks had lower census because of a holiday.~~ What does this pattern indicate?

Target word: interpret the run chart **Decides:** 6 or more consecutive points on one side of the median.

A	Camouflage	The shift rule is met regardless of those two weeks.
B	Correct	Seven points in a row below the median meets the shift rule.
C	Confusion set	A trend is 5 or more points steadily rising or falling.
D	Absolute language	A signal shows non-random change, not proof of cause.

7. Answer A Concept 4 • transfer

Which run chart pattern meets the rule for a trend?

Target word: reverse: pattern from rule **Decides:** 5 or more consecutive points all rising or all falling.

A	Correct	Five points steadily falling is a trend.
B	Confusion set	That is an astronomical point, a different signal.
C	Wrong stem number	Too few points for a trend.
D	Adjacent category	That is a shift, not a trend.

8. Answer B Concept 5

A project team ~~at a teaching hospital~~ compares average ED door-to-provider time in minutes for 200 patients before a triage change with a different 200 patients after it. ~~The team's statistician is a former high school math teacher.~~ Which analysis fits best?

Target word: choose a test **Decides:** Continuous outcome (minutes), two independent groups.

A	Confusion set	Chi-square compares category counts, not means.
B	Correct	Compares the means of a continuous measure between two independent groups.
C	Dropped element	Skips any check of whether the difference exceeds chance.
D	Adjacent category	Paired tests need the same subjects measured twice; these are different patients.

9. Answer A Concept 5 • transfer

A team wants to know whether the proportion of patients readmitted within 30 days (yes or no) differs between two discharge programs. Which analysis fits?

Target word: reverse: categorical outcome **Decides:** Yes/no outcome across two groups.

A	Correct	Compares proportions of a categorical outcome between groups.
B	Adjacent category	Correlation measures association between two continuous variables.
C	Inverted operation	Used for means of continuous data.
D	Confusion set	Run charts show change over time, not a two-group comparison.

10. Answer C Concept 6

To measure hand hygiene, a unit counts how many sanitizer pumps are used each shift. Two auditors ~~who both trained at the same university~~ agree on the counts 98% of the time. Staff admit some pumps are used to clean equipment. What is the best description of this measure?

Target word: reliability vs. validity **Decides:** Consistent counts, but pumps do not equal hand hygiene events.

A	Inverted operation	Reliability is necessary but not sufficient for validity.
B	Confusion set	Agreement is high, so reliability is not the problem.
C	Correct	Consistent between raters, but it does not accurately measure hand hygiene.
D	Camouflage	Shared training explains agreement, not accuracy.

11. Answer C Concept 7

A project team measuring time to first antibiotic for sepsis asks IT for a data extract. The team lead, ~~who has a new office near the cafeteria~~, requests every patient's full chart, including Social Security numbers and home addresses, "in case we need it later." What is the best response?

Target word: data request scope **Decides:** Only data needed for the measure should be pulled.

A	Camouflage	Location does not fix an over-broad request.
B	Dropped element	A permitted purpose still requires minimum necessary data.
C	Correct	The minimum necessary standard applies to quality improvement data use.
D	Absolute language	QI is an allowed health care operation; the issue is scope.

12. Answer D Concept 8

A central-line infection project on an ICU ~~with a new family lounge~~ will report its CLABSI rate quarterly. Because infections are rare, the nurse leader wants a weekly signal that the project is on track. ~~The ICU has 20 beds.~~ Which indicator best serves that purpose?

Target word: leading indicator **Decides:** Needs a frequent, predictive process measure.

A	Inverted operation	A lagging outcome measure that is too slow for weekly steering.
B	Camouflage	A fixed structure fact, not a signal of progress.
C	Adjacent category	Another lagging outcome, even slower.
D	Correct	Bundle compliance is a leading indicator that predicts infections and changes quickly.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Structure, process, and outcome measures	Structure = resources and setting (staffing, equipment); process = what is done and how reliably (percent of patients who received a step); outcome = the effect on patient health (falls, infections, mortality).	Calling a compliance percentage an outcome measure.	Whether the metric counts a resource, an action taken, or a patient result.
2	Balancing measures	Balancing measures watch for unintended consequences elsewhere in the system caused by the change (e.g., readmissions when length of stay drops).	Tracking only the target metric and missing harm created elsewhere.	A change that could push a problem to another part of the system.
3	Building a rate-based KPI	A KPI needs a clear numerator, denominator, and multiplier (e.g., falls / patient days × 1,000) so results can be compared across time and units.	Reporting raw counts that rise and fall with volume.	Event counts plus patient days, admissions, or device days.
4	Run chart rules	On a run chart, a shift is 6 or more consecutive points on one side of the median, and a trend is 5 or more consecutive points all going up or all going down; either signals non-random change.	Declaring success from one or two good data points.	A sequence of points described relative to the median.
5	Choosing an analysis by data type	Compare proportions or counts in categories with a chi-square test; compare the means of a continuous measure between two independent groups with a t-test.	Using a t-test on yes/no outcomes or a chi-square on averages.	Whether the outcome is a category (yes/no) or a continuous number.
6	Reliability vs. validity	Reliability is consistency (the same result each time or between raters); validity is accuracy (measuring what it is meant to measure). A measure can be reliable but not valid.	Assuming high inter-rater agreement proves the measure is valid.	A tool that gives consistent results but may capture the wrong thing.
7	Minimum necessary and de-identified data	Quality improvement data collection should use only the minimum protected health information needed, and de-identified or limited data sets when they are enough.	Pulling entire records “just in case.”	A data request that includes identifiers the measure does not need.

#	Concept	Rule	Trap	Tell
8	Leading vs. lagging indicators	Lagging indicators measure end results after the fact (infection rates); leading indicators track earlier process behaviors that predict those results (bundle compliance) and allow course correction sooner.	Relying only on outcome rates that change too slowly to steer the project.	A need for early feedback during implementation.

Course scope (public description): Evidence-based measures for evaluating healthcare improvements is an essential component of the planning phase of the healthcare improvement project. In this course, students will determine key performance indicators and metrics used to determine the success of a healthcare improvement project (HIP). The student will develop collaborative partnerships and build consensus with stakeholders to determine how specific data will be collected, managed, and analyzed. This is also an opportunity to discuss data issues and technologies needed for the project. To accomplish this phase, students will also determine the parameters, procedures, and technologies needed for data collection, management, analysis, and reporting. This course is eligible for an In Progress grade. Please see the Grading Scale Policy for more information.